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Ising Model Simulation with a Quantum Computer

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Trabajo final de Entrelazamiento Cuántico e Información.

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January 29, 2024

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Abstract. In this report we show our results for simulating the magnetic behaviour of a spin chain using the Ising Model for four fermions via exact simulations with the package of python "qiskit" and real measures with the quantum computer of IBM "Kyoto". To carry out this task we have developed a quantum circuit of four qubits based on previous works in this area, in which stands out Cirac et al Ref. [7], Merino et al Ref. [5], and above all Alba Cervera-Lierta's publication Ref. [2]. The aim of this circuit is to transform the states of the computational basis into the eigenstates of the Ising Hamiltonian. With this tool we have calculated first with qiskit the exact result from the expected values of some Pauli operators and with certain dependencies such as the external magnetic field (λ), time (t) or temperature (T) (this are: $\langle\sigma_z(\lambda)\rangle$, $\langle\sigma_z(t)\rangle$, $\langle\sigma_x(t)\rangle$, $\langle\mathcal{H}\rangle$ and $\langle\sigma_z(T)\rangle$). Given the exact simulations we compare them with the measures from the IBM quantum computer, and extending the works we have been inspired by, we propose a noise model to explain the differences observed between the theoretical values of transverse magnetization and the quantum simulation prediction, and we finally double-test our model using other experimental characteristics of the device.

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1 Introduction

For predicting any observable of a quantum physical system it is helpful to have the whole spectrum of energies of the system. However, the number of dimensions of the Hilbert space of the system grows exponentially with the number of qubits ($\propto 2^n$), and an exact diagonalization of the Hamiltonian becomes computationally very hard for a few tens of qubits. In this context, we ask ourselves if, maybe, Quantum Computers will be able to exactly simulate in a more efficient way quantum many-body systems, as Feynman aimed when he firstly proposed Quantum Computing [3].

# of sites	# of states	Hamiltonian size in memory
4	16	2048 B
9	512	2 MB
16	65536	34 GB
25	33554432	9 PB
36	6.872e10	40 ZB

Table 1: Hamiltonian size in memory of Ising Hamiltonian for different number of qubits.

Our objective in this project is to reproduce an algorithm that permits us simulate the one-dimensional Ising Model for the special case of four qubits.

In section 2 we will see how to create the circuit needed to measure through a quatum computer the eigenstates of this Ising chain, specifying step by step the gates needed to go from the computational basis to the Ising Hamiltonian basis.

In section 3 we show some of the results we obtained from this circuit, by means of an IBM quantum computer (Kyoto) and the python library qiskit, such as the dependency of the transverse magnetization $\langle\sigma_z\rangle$ with the external magnetic field parameterized by λ in subsection 3.1. We also propose a noise model to try to understand the difference between exact theoretical values and those predicted by Kyoto[1]. In subsection 3.2 we study the dependence of $\langle\sigma_z\rangle$ with time "t". And, to show more possibilities that this algorithm can offer, we also checked the results for $\langle\sigma_x(t)\rangle$ in subsection 3.3 and we finally perform a thermal simulation to compute the magnetization dependence with temperature "T", $\langle\sigma_z(T)\rangle$, in subsection 3.4.

2 Circuit construction

For the development of a quantum algorithm that diagonalizes the Ising Model Hamiltonian we are going to need a certain combination of gates that reproduces the already proposed diagonalization of the Transverse Field Ising Model [1] in form of quantum circuit. Initially, in the spin basis we have a Hamiltonian[2]:

$$\mathcal{H}_{ising} = \sum_{i=1}^{n=4} \left(\sigma_x^i \sigma_x^{i+1} + \lambda\sigma_z^i\right) + \sigma_y^1\sigma_z^2\sigma_z^3\sigma_y^4, \tag{2.1}$$

where the last term is implemented to cancel a frontier term that emerges during the diagonalization, as we are taking periodic boundary conditions [1]. Our basis is the spin basis B, with eigenvectors $|\uparrow\uparrow\downarrow\uparrow\rangle, |\uparrow\uparrow\downarrow\downarrow\rangle$ ... which will get corresponded with the computational basis of the qubits with the identification $|0\rangle \to |\uparrow\rangle$ and $|1\rangle \to |\downarrow\rangle$. As we appreciate in this basis our Hamiltonian is not diagonalized. We proceed to apply the required transformations for this task.

2.1 Step 1: Jordan Wigner transformation

This transformation expresses the system's degrees of freedom as fermionic modes. A fermionic mode is created in the system when one spin flips from its fundamental state to the excited one. The associated operators of destruction and creation of this modes must respect the anticommutation relations of fermions [1]:

$$\{c_i, c_j^\dagger\} = \delta_{ij}\,, \quad \{c_i, c_j\} = 0. \tag{2.2}$$

The transformation can be performed like this:

$$c_j = \left(\prod_{l<j} \sigma_z^l\right) \otimes \frac{\sigma_x^j + i\sigma_y^j}{2} \tag{2.3}$$

[1]It is worth remarking that what we performed in the quantum computer Kyoto is an exact simulation of the Ising Model. Still, quantum noise and errors will make the values predicted by Kyoto deviate from the exact analytically predicted values. However, qiskit will provide us the possibility to simulate the circuit as if there was no noise or errors, so we can obtain the exact prediction.

[2]Where $\sigma_{(i)}^z$ goes for: $\sigma_{1z} = \sigma_z \otimes I \otimes I \otimes I, \quad \sigma_{2z} = I \otimes \sigma_z \otimes I \otimes I$ and so on.

So we then rewrite the Hamiltonian in terms of these operators:

$$\mathcal{H}_{ising} \to \mathcal{H}^c_{ising} = \frac{1}{2}\sum_i^n \left(c_i^\dagger c_{i+1} + c_{i+1}^\dagger c_{i+1} + c_i c_i + c_{i+1}^\dagger c_i^\dagger + \lambda c_i^\dagger c_i\right) \tag{2.4}$$

We can express the wave-function of the system in the following two ways:

$$|\psi\rangle = \sum_{i_1,i_2,i_3,i_4=0,1} \psi_{i_1 i_2 i_3 i_4} |i_1 i_2 i_3 i_4\rangle = \sum_{i_1,i_2,i_3,i_4=0,1} \psi_{i_1 i_2 i_3 i_4} \left(c_1^\dagger\right)^{i_1}\left(c_2^\dagger\right)^{i_2}\left(c_3^\dagger\right)^{i_3}\left(c_4^\dagger\right)^{i_4} |i_1 i_2 i_3 i_4\rangle \tag{2.5}$$

As we see, in the computational basis B and in the new basis B^a the coefficients of the wave-function are exactly the same. This means that we do not need any quantum gate to perform this change of basis.

We must, however, take into account that we are dealing with fermions so any time we exchange two qubits in state $|1\rangle$ will mean that we are exchanging two fermionic modes, and that the wave-function is anti-symmetric under that operation.

Then for performing a change of spins we use the FSWAP gate [7]:

$$\text{FSWAP} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix} \tag{2.6}$$

2.2 Step 2: Fourier transform

The next step is to perform a Fourier transformation to go to momentum-space. Redefining our degrees of freedom in terms of operators of creation/annihilation in momentum space:

$$b_k^\dagger = \frac{1}{\sqrt{N}}\sum_{j=1}^n exp\left(i\frac{2\pi j}{n}k\right) c_i^\dagger, \quad k = -\frac{n}{2}+1, ..., \frac{n}{2} \tag{2.7}$$

our Hamiltonian gets the following form:

$$\mathcal{H}^b_{ising} = \sum_{k=-\frac{n}{2}+1}^{\frac{n}{2}} \left[2\left(\lambda - cos\left(\frac{2\pi k}{n}\right)\right) b_k^\dagger b_k + isin\left(\frac{2\pi k}{n}\right)\left(b_k^\dagger b_{-k}^\dagger + b_k b_{-k}\right)\right], \tag{2.8}$$

as we can see, it is not diagonalized yet because we have a coupling between opposite sign momenta.

The way to implement the Fourier transform is to divide the transformation into odd and even modes:

$$\sum_{j=0}^{n-1} exp\left(i\frac{2\pi}{n}k\right) c_j^\dagger = \sum_{j'=0}^{\frac{n}{2}-1} exp\left(i\frac{2\pi j'}{n/2}k\right) c_{2j'}^\dagger + exp\left(i\frac{2\pi k}{n}\right)\sum_{j'=0}^{\frac{n}{2}-1} exp\left(i\frac{2\pi j'}{n/2}k\right) c_{2j'+1}^\dagger \tag{2.9}$$

As we have an even number of qubits, four, we can perform the Fourier transform using 2-qubit gates which have this form[3]:

[3]Derivation in appendix.

$$\text{FT-gate} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & \frac{e^{2\pi ik/n}}{\sqrt{2}} & 0 \\ 0 & \frac{1}{\sqrt{2}} & \frac{-e^{2\pi ik/n}}{\sqrt{2}} & 0 \\ 0 & 0 & 0 & -e^{2\pi ik/n} \end{bmatrix} \tag{2.10}$$

2.3 Step 3: Bogoliubov transform

The Bogoliubov transformation is an isomorphism of the algebra of canonical commutations and anticommutations, commonly used in quantum physics to diagonalize Hamiltonians.

In our case we will use it to decouple the opposite momenta sign operators, which is the final transformation we need to apply to arrive to the eigenstates basis of the diagonal Hamiltonian which has the form:

$$\text{B-gate} = \begin{bmatrix} cos\left(\frac{\theta_k}{2}\right) & 0 & 0 & isin\left(\frac{\theta_k}{2}\right) \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ isin\left(\frac{\theta_k}{2}\right) & 0 & 0 & cos\left(\frac{\theta_k}{2}\right) \end{bmatrix} \tag{2.11}$$

This leaves finally the Hamiltonian with the form:

$$\mathcal{H}^a_{ising} = \sum_{k=-n/2+1}^{n/2} w_k a^\dagger_k a_k \tag{2.12}$$

Where [7]:

$$a_k = cos\left(\frac{\theta_k}{2}\right) b_k - isin\left(\frac{\theta_k}{2}\right) b^\dagger_{-k}; \tag{2.13}$$

$$\theta_k = arcos\left(\frac{-\lambda + cos\left(\frac{2\pi k}{n}\right)}{\sqrt{\left(\lambda - cos\left(\frac{2\pi k}{n}\right)\right)^2 + sin^2\left(\frac{2\pi k}{n}\right)}}\right); \tag{2.14}$$

$$w_k = \sqrt{\left(\lambda - cos\left(\frac{2\pi k}{n}\right)\right)^2 + sin^2\left(\frac{2\pi k}{n}\right)} \tag{2.15}$$

Hence, the Hamiltonian has now the form of a sum of non-interacting terms and its spectrum can be represented by the computational basis as we will see in the following subsection.

2.4 Circuit usage

Finally we diagonalized our Hamiltonian:

$$\mathcal{H}^a_{ising} = U^\dagger_{dis} \mathcal{H}_{ising} U_{dis}, \tag{2.16}$$

by the creation of an algorithm that reproduces the unitary operation U_{dis}, which is a change of basis from B to B^a. However, recall that our purpose is having the spectrum of eigenvectors of the Hamiltonian. More specifically, we would like to have them in the basis

B, as it is the basis in which we have written all the possible operators with which we want to compute.

If we take advantage of the fact that as $\mathcal{H}^a$ is a diagonalized or non-interacting Hamiltonian, we are able to reach the spectrum in the basis B in the following way:

$$\mathcal{H}^{a\prime} = \sum_i \epsilon_i \sigma_z^i, \tag{2.17}$$

where ϵ_i are the eigenvalues. The eigenvectors of this Hamiltonian are then the 16 vectors computational basis: $\{|0001\rangle, |0011\rangle ...\}$.

If we have an eigenvector of $\mathcal{H}^{a'}$, for example $|0001\rangle$ and we apply the inverse change of basis, $U_{dis}^\dagger$, we will obtain the same eigenvector but in the basis B. This implies that initializing in all the 16 vectors of the computational basis or B^a we can obtain the whole spectrum in the basis B.

So this circuit gave us the possibility of, without analytically diagonalizing the Hamiltonian, obtaining then the exact analytical spectrum. We can finally make predictions about this system, and our purpose now is to get different results to illustrate some of the possibilities we have.

2.5 Topology of the circuit

The circuit in order to perform the unitary transformation $U_{dis}^\dagger$ is shown bellow:

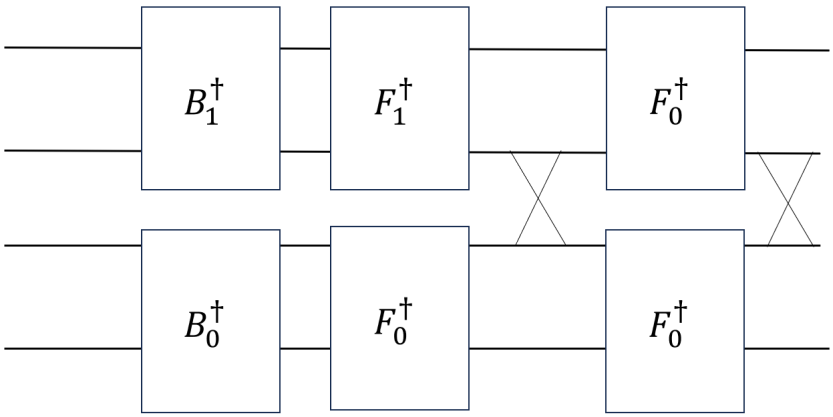


Figure 1: Topology of the quantum circuit to perform the operation $U_{dis}^\dagger$. The crossed lines represent FSWAP transformations.

The circuit separate the modes into even and odd and perform the transformations needed separately. There are infinitely more ways to construct the circuit, this is inspired in Alba's publication Ref.[2], but not exactly the same, to see another way consult Ref. [7].

3 Results and discussion

In this section we will try to make the most of this circuit by checking the results of some analytically predicted magnitudes such as the transverse magnetization in order to reveal the usefulness of this quantum computing algorithm, and we will also study the quantum noise that limits the device to perform better.

3.1 Transverse magnetization $\langle \sigma_z \rangle$ of the ground state as a function of the control parameter λ

Now, that we have the whole spectrum of eigenvalues and eigenvectors of $\mathcal{H}$, our first task is to calculate the transverse magnetization of the ground state as a function of λ. We need to remark that the ground state eigenstate changes depending on the region of λ, as the magnetic field induces quantum quenching [4], that we are studying [2]:

$$|gs\rangle = |0001\rangle \text{ if } \lambda < 1 \text{ and } |0000\rangle \text{ if } \lambda > 1 \tag{3.1}$$

So, for different values of λ we have to initialize the circuit in the corresponding eigenstate. Then we have to run the circuit in IBM Quantum platform enough times to have statistics. In our case we chose "IBM-Kyoto" to do the measures.

Then we will obtain $\langle \sigma_z(\lambda) \rangle$ by running the circuit enough times for each λ to have a statistical significant sample.

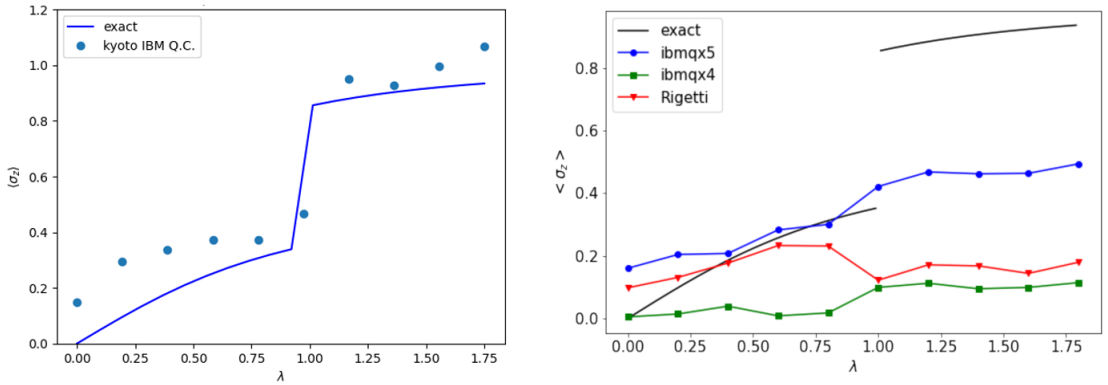


Figure 2: Transverse magnetization $\langle \sigma_z \rangle$ as a function of the control parameter λ. At left is plotted our quantum simulation with IBM-Kyoto and at the right Alba Cervera's 2018 simulation from Ref. [2] with other quantum devices no longer available .

What we observe in Figure 2 is that the quantum simulation developed in IBM-Kyoto captures the pattern of the exact theoretical model dependence with λ but we can appreciate a certain bias of the quantum simulation to bigger values of transverse magnetization.

It is important to compare our simulation with the results obtained by Alba Cervera-Lierta, Ref. [2]. What we appreciate just by observation is that our quantum simulation performs significantly better. We conclude then that the development of IBM quantum devices during these last six years is really notorious. In our particular case, IBM-Kyoto seems to show a great improvement with respect to their predecessors for this quantum simulation.

Despite the nice improvement, the difference between the theoretical curve and our quantum simulation is evident. We need to remember that when performing a quantum algorithm the number of shots, or times we run the circuit, can be determinant to get predictions comparable with theory. Having a bigger sample of measures will make calculations of mean values of any observable more accurate.

[4] For more information see Appendix B of Ref. [1]

It would be expected that increasing the statistical sample will improve the performance of the simulation. For the quantum simulation of Figure 2, we performed 1024 shots for each estimation of $\langle \sigma_z \rangle$ associated to each λ. We will now duplicate the sample size to 2048 shots and compare both cases.

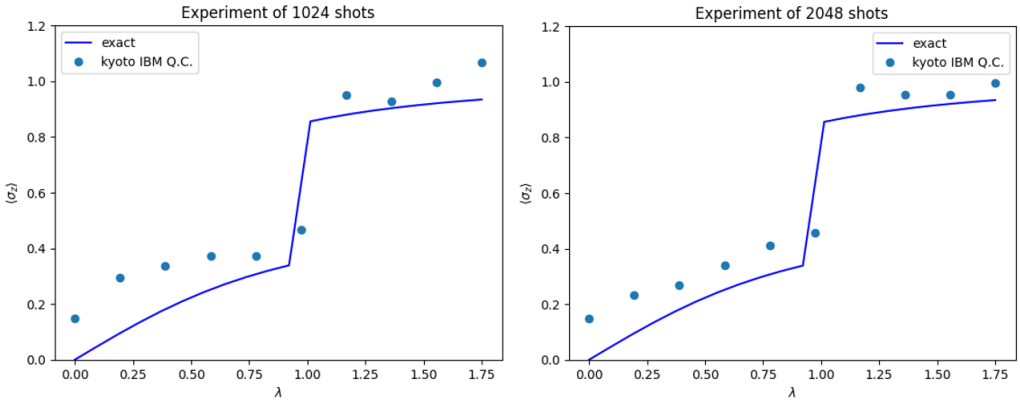


Figure 3: At the left we plotted again the transverse magnetization σ_z as a function of λ for the 1024 shots quantum simulation, and at the right side for 2048 shots.

After repeating the experiment with the double of statistical sample we observe that the quantum simulation gets closer to the theoretical curve, but the bias to bigger values of the transverse magnetization is still present. Of course, we could still repeat the simulation with even greater number of shots but we have the IBM compilation time limitation, so maybe we should try another approach to try to explain it.

3.1.1 Quantum noise model creation

Quantum noise is a very important factor that we yet have not considered, and there are lot of different sources, such as coupling with environment or bad implementation of gates, that can make the algorithm work differently as one would expect ideally [6].

We created a model of quantum noise so we could understand what kind of quantum noise are we suffering during our simulation. IBM Quantum Experience offers us for each quantum device the median quantum errors of different kinds [4]. More specifically for our choice, IBM-Kyoto:

$$\begin{aligned} \text{median } ecr \text{ error} &= 8.2 \times 10^{-3}; \\ \text{median } S_x \text{ error} &= 2.483\,10^{-4}; \\ \text{median readout error} &= 1.51 \times 10^{-2} \end{aligned}$$

We therefore created a noise model we called "Kyoto noise model" with those errors, simulated the circuit and compared the results with the experimental points. However, as we see in Figure 4, this noise model does not capture the experimental pattern.

Then, we tried to improve the noise model. As we have already said the states $|0/1\rangle$ of a qubit simulates the states $|\uparrow / \downarrow\rangle$ of an Ising model spin. Usually $|0\rangle$ denotes the fundamental and $|1\rangle$ the excited state of a qubit. So there is a no-null probability of qubits which have an state $|1\rangle$ to decay to $|0\rangle$. We realized that this would imply to have more counting of $|0\rangle$

states for the final state of the system than expected initially, and as these states correspond to $|\uparrow\rangle$ states of the spins simulated, we will measure larger values of σ_z.

To see if our hypothesis is right we propose "Kyoto noise model 2.0". It consists in adding a readout error due to a probability $p_d = 0.14$ of decay of single qubits $|1\rangle \to |0\rangle$. The way to implement it is the following:

We first ran the experiment which gave us $N_{0000}, N_{1000}, N_{0100}, N_{0100}$ and so on. We updated these values in the following way:

$$\begin{aligned}
N'_{1111} &= N_{1111} \times \left(1 - p_d - p_d^2 - p_d^3 - p_d^4\right) \\
N'_{1x3} &= N_{1x3} \times \left(1 - p_d - p_d^2 - p_d^3\right) + N_{1111} \times p_d \\
N'_{1x2} &= N_{1x2} \times \left(1 - p_d - p_d^2\right) + N_{1x3} \times p_d + N_{1111} \times p_d^2 \\
N'_{1x1} &= N_{1x1} \times (1 - p_d) + N_{1x2} \times p_d + N_{1x3} \times p_d^2 + N_{1111} \times p_d^3 \\
N'_{0000} &= N_{0000} + N_{1x1} \times p_d + N_{1x2} \times p_d^2 + N_{1x3} \times p_d^3 + N_{1111} \times p_d^4,
\end{aligned}$$

where we denote N_{1xn} as the sum of all the times the states with "n " ones were the final state of the circuit. The model assumes that the different qubits are not correlated, and then the decay of the different qubits of the systems are independent events. That is why, for instance, the number of states that initially were $|0000\rangle$ and de-excited to $|1111\rangle$ are $N_{1111} \times p_d^4$ as the decay of four qubits has a probability p_d^4 of happening.

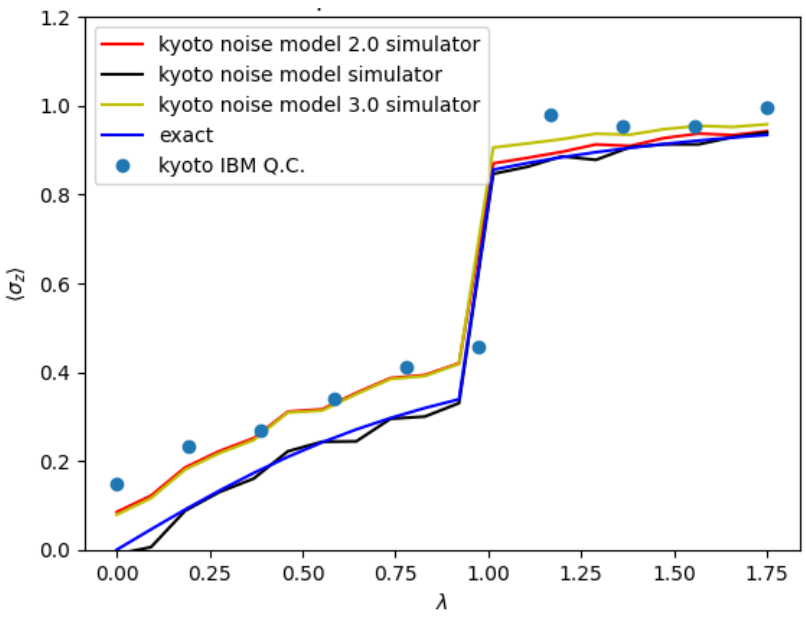


Figure 4: Mean transverse magnetization σ_z as a function of λ. We plot the analytical prediction (blue), the experimental points obtained by IBM-Kyoto, and the diverse noise model simulation predictions.

We observe that the "Kyoto noise model 2.0" adjust much better to the data in comparison with the initial model "Kyoto noise model" for $\lambda < 1$. This could mean that, actually, these decays of the qubits play an important role in the quantum simulation.

However, for $\lambda > 1$ the noise model 2.0 gets worse. In order to find a model which adjust also to this regime of λ we postulate that it exists a probability $p_{trans} = 0.25$ of two qubit decay $|11\rangle \rightarrow |00\rangle$. Despite of adjusting the data better, there is not much physical motivation for this kind of transition to be so likely, so we may not consider it as a valid noise model.

3.1.2 Testing the decay hypothesis: Thermal relaxation times.

There is output data that IBM brings us to verify if the decay during the measurement time has such an impact in the simulation. More specifically, thermal relaxation time is defined as the time needed for a qubit to decay from the excited state $|1\rangle$ to the ground state $|0\rangle$ [4].

Then, in a certain time "t" the population of a state initially excited ($|\psi\,(t=0)\rangle = |1\rangle$) is given by [8]:

$$\text{Population of}\,|1\rangle\,(t) = \text{Population of}\,|1\rangle\,(t=0)\,exp\,(t/T_1) \tag{3.2}$$

$$\rightarrow \text{Probability of decay}\,(t) = (1 - exp\,(t/T_1)) \tag{3.3}$$

IBM Quantum Experience gives us the data of the thermal relaxation times of each qubit, and also the readout time, this means that we can check if the probability of decay given by eq. 3.3 is the same as the one we had hypothesized in our noise model, $p_d = 0.14$.

The data given by IBM Quantum experience, is respectively for each qubit: $T_1 =$ 172.04, 110.17, 205.72 and 124.94 μs. The readout time for every qubit is 14 μs. This end up giving a probability of decay during readout:

$$\text{Probability of decay during readout} \approx 0.01 \tag{3.4}$$

We appreciate a clear contradiction between the probability we had as a hypothesis and the one calculated from experimental data. We can conclude then that, even thought our noise model fitted reasonably well the quantum simulation curve of $\langle\sigma_z\rangle$ as a function of λ, the model is either underestimating or not taking into account some noise relevant sources and overestimating the probability of decay during readout time. It is worth to remark that the effect of decay during readout time is not negligible, and that perhaps a model that implements this probability of decay during the whole circuit implementation gets closer to explain the quantum noise of the device.

3.2 Temporal evolution of $\langle\sigma_z\rangle$ for different λ

Once we have developed our circuit to obtain the eigenstates of the Ising Hamiltonian from section 2 we are able to implement some new features to it in order to obtain new observables related with the magnetization of the chain. Here we see the temporal evolution $\langle\sigma_z\rangle$ for the initial state $|\psi_0\rangle = |\uparrow\uparrow\uparrow\uparrow\rangle$, which corresponds to $|0000\rangle$. Analytically we obtained the result by applying the operator $U^\dagger_{dis}$ representing the circuit and the new operator involved $U(t) = e^{-it\mathcal{H}}$ for temporal evolution to this initial state:

$$|\psi(t)\rangle = U^\dagger_{dis}U(t)\,|\psi_0\rangle = (cos(\phi)\,|00\rangle + isin(\phi)e^{it4\sqrt{\lambda^2+1}}\,|11\rangle) \otimes |00\rangle \tag{3.5}$$

Where we know the eigenvalue [5] of the Hamiltonian for this state from Ref.[2], ϕ goes for the angle used to in the Fourier Transform gates so for each value of "k " it would have a

[5] Albeit we have used this analytical value of energy, from this point onwards we will only obtain this parameter using the quantum simulator of qiskit.

different form. Having defined the state dependent of "t " now we are interested in obtaining $\langle\sigma_z(t)\rangle$. We must consider that for one qubit, the Pauli matrix takes the form:

$$\sigma_z(1 \text{ qubit}) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{3.6}$$

And therefore, for four qubits is:

$$\sigma_z = \frac{1}{4}\sum_i^4 \sigma_z^i \tag{3.7}$$

Applying the final state to this operator we obtain the expected value of $\langle\sigma_z\rangle(t)$:

$$\langle\sigma_z\rangle(t) = \langle\psi(t)|\,\sigma_z\,|\psi(t)\rangle = \frac{1 + 2\lambda^2 + cos(4t\sqrt{1+\lambda^2})}{2 + 2\lambda^2} \tag{3.8}$$

Going back to the circuit, the temporal evolution is implemented through the gates showed in Figure 5.

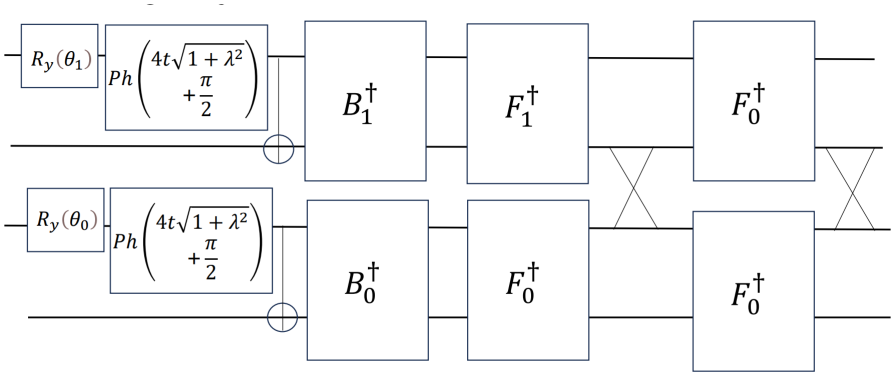


Figure 5: From the original circuit (see Figure 1) we first add a rotation "Y" gate to include the angle ϕ from equation 3.5, the next gate adds the temporal evolution phase with a $\frac{\pi}{2}$ factor associated with the imaginary unit "i" appearing in this same equation. Finally, the CNOT gate apply this operations to the following qubit.

This circuit was sent to the IBM Quantum Computer Kyoto for different times and different λ, obtaining the results showed in Figure 6. In this figure we see that $\langle\sigma_z(t)\rangle$ follows the same behaviour than expected by eq. 3.8 and all three cases oscillate with the same frequency than the exact simulation. Even so, there exists some difference between the simulation and the measures, that may be produced by the quantum noise of the computer. In this case in contradistinction with the Figure 9 we see that points tend to be beneath the expected value, above all for high σ_z. This is because looking at eq. 3.5 we first see that as the state $|1100\rangle$ doesn't contribute to the mean value of σ_z, we just care about the probability of obtaining the state $|0000\rangle$ that is the one that can give a bigger contribution to the expected value of this observable, any deviation of the probability of obtaining the four spins up because of the computational system errors would produce a different final state

with a smaller contribution. This bias is a sign that the decay error is not the only significant in the quantum noise, as we hypothesized in subsection 3.1, of the device as it would produce the bias towards smaller values of σ_z.

This kind of problem is also seen for Alba Cervera's work in this same figure, but in a more accentuated way because back in 2018 the quantum machines were less developed and therefore less accurate. Thus, this results shows as well how much IBM computers have improved since then.

3.3 Temporal evolution of $\langle\sigma_x\rangle$ for $\lambda = 0.9$

To make sure that the circuit was correct in measuring $\langle\sigma_z\rangle$ we found also interesting to measure $\langle\sigma_x\rangle$ and see if it also meets our expectations.

The operator σ_x applied to a certain eigenstate switches it to the other, in the case of the computational basis, this Pauli matrix changes a 0 for a 1 and vice versa. This is because of its form:

$$\sigma_x(1 \text{ qubit}) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \tag{3.9}$$

Hence, our final states would changes in this way:

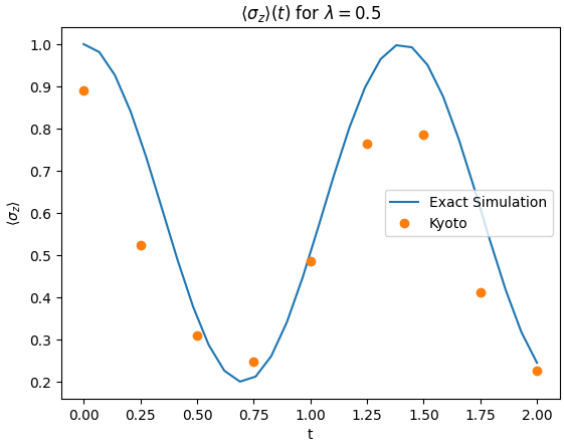
$$\sigma_x |0000\rangle = |1111\rangle\,; \quad \sigma_x |1100\rangle = |0011\rangle \tag{3.10}$$

$$\sigma_x |\psi(t)\rangle = (cos(\phi)\,|11\rangle + isin(\phi)e^{it4\sqrt{\lambda^2+1}}\,|00\rangle) \otimes |11\rangle \tag{3.11}$$

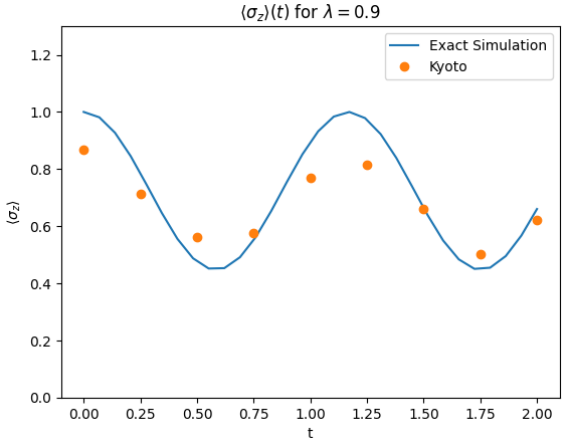
As the elements of the computational basis are orthonormal, the final expected value of σ_x we get is:

$$\langle\sigma_x\rangle = \langle\psi(t)|\,\sigma_x\,|\psi(t)\rangle = 0 \tag{3.12}$$

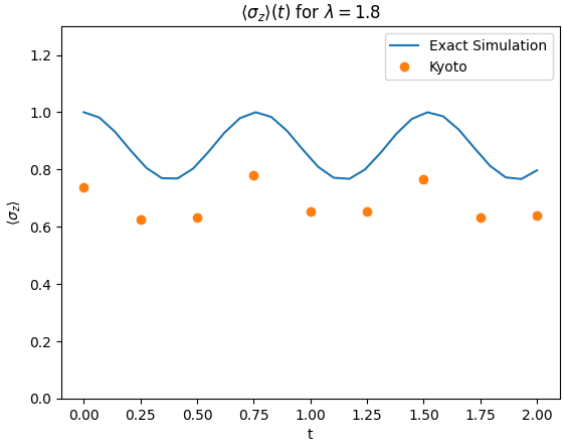
To measure this observable we just need to change the basis of the final states from σ_z to σ_x introducing a Hadamard gate at all the qubits before measuring them from the circuit of Figure 5. The results of this measures are shown in Figure 7. In this figure we see that both the simulation and the measures follow the result of eq. 3.12 in spite of some little error from the quantum computer Kyoto which seems to be completely random and produced by the noise of the machine. This results let us be even more sure that our circuit is correctly constructed.



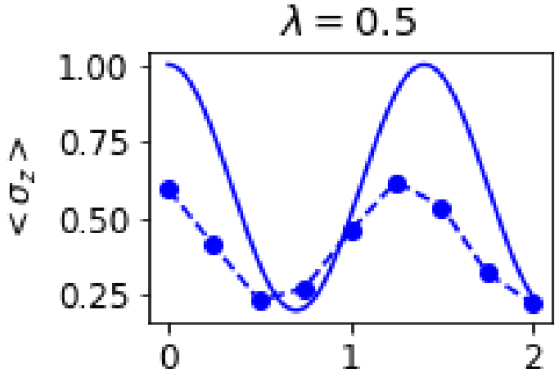
(a) Our results for $\lambda = 0.5$



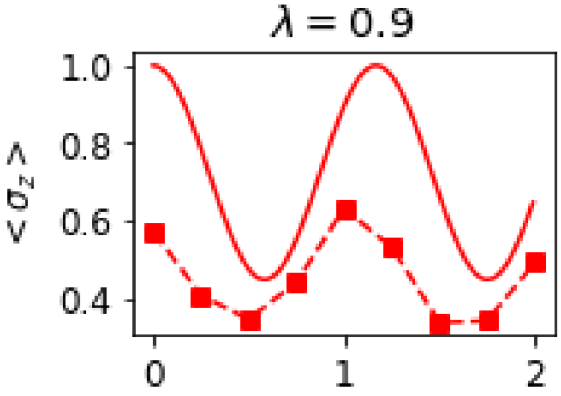
(b) Our results for $\lambda = 0.9$



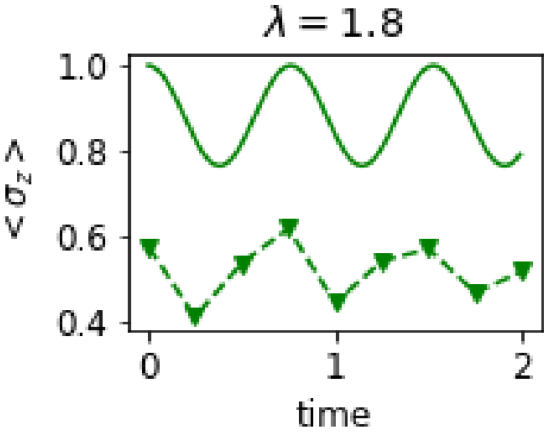
(c) Our results for $\lambda = 1.8$



(d) Alba Cervera's result for $\lambda = 0.5$



(e) Alba Cervera's result for $\lambda = 0.9$



(f) Alba Cervera's result for $\lambda = 1.8$

Figure 6: Transverse magnetization $\langle \sigma_z \rangle$ as a function of time for 3 different values of the control parameter λ. At left is plotted our our exact simulation in blue and the the IBM-Kyoto quantum computer measures are represented by the orange dots, whilst at right are is Alba Cervera's 2018 Ref.[2] results. The exact function is represented by the filled line and the measures from a former quantum IBM computer (2018) by the dots joined by the dashed line .

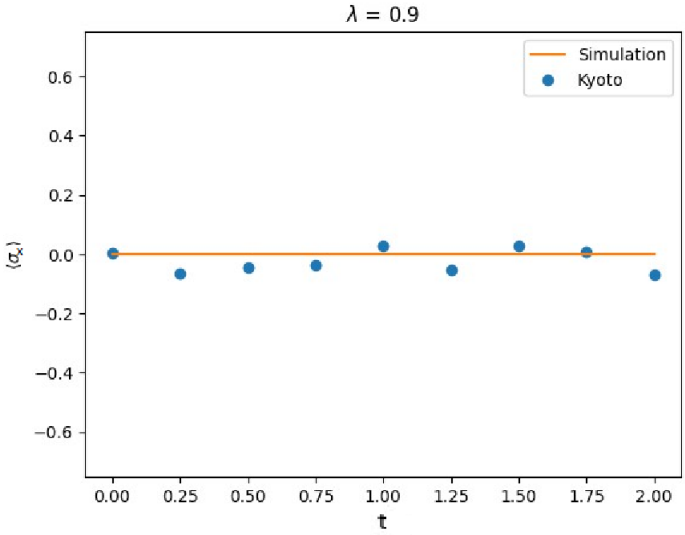


Figure 7: Expected value of σ_x as a function of time for $\lambda = 0.9$. In orange we find the exact simulation made with qiskit of the circuit that for all times gives a expected value of 0 in accordance to eq. 3.12 whilst in orange are the dots that represents the measures from the IBM quantum computer Kyoto.

3.4 Thermal simulation: $\langle \sigma_z \rangle$ in terms of temperature and λ

For the previous subsections we have confirmed that the circuit enables us to measures the expected value of some simple observables as the Pauli matrices, but if we want to go further in the use of this circuit we might be interested in getting to know the eigenvalues of energy, which were present in the temporal evolution but there we just used two of them that were already calculated from previous works.

As we have argued before, by initializing in the 16 eigenstates of the computational basis we obtain the spectrum of eigenvalues of energy $|E_i\rangle$. Regarding the circuit, the only thing we add to Figure 1 is a "x-gate" for each "1" qubit for each initial state. Then to get the eigenvalues of energy is as easy as obtaining the mean values of $\mathcal{H}_{ising}$ for each eigenstate:

$$\langle E_i | \mathcal{H}_{ising} | E_i \rangle = \epsilon_i \tag{3.13}$$

Knowing the whole spectrum we can do a thermal simulation using Boltzmann distribution:

$$\rho(\beta) = \frac{e^{-\beta\mathcal{H}}}{Z} = \frac{1}{Z}\sum_i e^{-\beta\epsilon_i} |E_i\rangle \langle E_i|, \; Z = \sum_i e^{-\beta\epsilon_i}, \; \beta = \frac{1}{k_b T}, \tag{3.14}$$

where "Z" is the partition function and "ρ" is the density matrix.

Having the thermal distribution we can directly calculate any observable mean dependence with temperature "T" and the control parameter λ:

$$\langle \mathcal{O}(\beta) \rangle = Tr[\mathcal{O}\rho(\beta)] = \frac{1}{Z}\sum_i e^{\beta\epsilon_i} \langle E_i | \mathcal{O} | E_i \rangle \tag{3.15}$$

For example, we compute the expected value of the transverse magnetization σ_z as a function of the magnetic field represented by λ and the temperature T with:

$$\langle \sigma_z(\beta) \rangle = \frac{1}{Z} \sum_i e^{\beta \epsilon_i} \langle E_i | \sigma_z | E_i \rangle \tag{3.16}$$

The results of this simulation for a given interval of λ and T are represented in Figure 8. There we can see that for high temperatures and low external magnetic fields, the material remains antiferromagnetic but as λ increases and T decreases it starts a phase transition to paramagnetic, where the magnetic field is strong enough to counter the effect of the temperature that tends to maintain it in the antiferromagnetic state.

We then conclude that the simulation we performed gave physical coherent results.

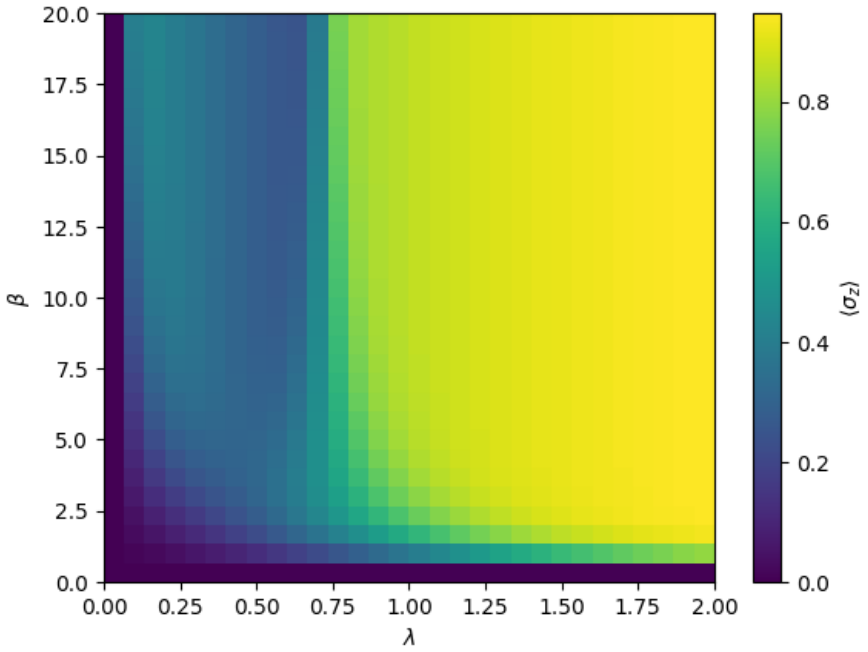


Figure 8: For a resolution of 30x30 frames we obtain a colour map in which the magnetic field parameter λ is represented in the x-axis and the thermal statistical factor "β"in the y-axis. The expected value of σ_z goes from 0 where the material is antiferromagnetic, i.e all spins have the same direction and are equally distributed in opposite senses, to paramagnetic, where it is paramagnetic and all spins are aligned with the external magnetic field.

4 Conclusions

In this work we developed an algorithm that allows us to obtain the eigenstates of the Ising Hamiltonian with transverse magnetization for four fermions. By means of this circuit we have made exact simulations with the computing system qiskit and measures with an IBM Quantum Computer (Kyoto) to get the expected value of several observables.

Comparing the results of the quantum computer with the exact solution, the circuit has proven to fulfill its purpose as we can appreciate a great accordance between them. Furthermore, the measures of the transverse magnetization $\langle\sigma_z(\lambda)\rangle$ and $\langle\sigma_z(t)\rangle$ seems to reveal a remarkable improvement in quantum computers' performance since the outcome of Alba Cervera's publication [2] in which we have inspired our work. Although the measures are more precise compared with those of previous works we still have some inaccuracies due to quantum noise.

In order to extend our work and find a simple solution to this problem we have developed a noise model, based mostly in a qubit decay probability, built to match the $\langle\sigma_z(\lambda)\rangle$ results and that, although it fits considerably well the points with the exact solution, it can be considered an ad hoc solution for a specific case that may not be generalized in some other instances. To check our model we computed the decay probability with thermal relaxation times of Kyoto's qubits, and we observe that they do not match with our hypothesis. Consequently we can infer that quantum noise modeling may be a great deal for the present and future development of quantum computers due to the multiple sources of decoherence in them.

We have also found interesting to explore some additional methods to make sure that the computer's circuit was right and we weren't getting good results by chance. The system we though to accomplish this job was the measure of $\langle\sigma_x(t)\rangle$ which was expected to be zero, thus we used the proximity of points to this value as a calibrator of our circuit's accuracy.

Finally we decided to test the capability of this algorithm with a thermal simulation, computing $\langle\sigma_z(T)\rangle$ by means of the eigenvalues of the Hamiltonian, which gave us some physically coherent results.

In conclusion, even though the progress in this area has been huge, there is still a long way to achieve a substantial accuracy in quantum computers so they can become an efficient tool for quantum simulations. But despite its difficulties, it has proven to have the potential to unseat classical computing in tasks such as those in which statistics and physics are present in which it stands out the case of the Ising Model we have been treating, which needs a negligible time in comparison with, for instance, a Montecarlo algorithm, and with the advantages of obtaining the eigenstates of a quantum system without needing the use of computational diagonalization of a matrix, as we mentioned in the Introduction 1. This could mean a great progress in, for example, the areas of mechanical statistics and condensed matter.

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5 Appendix

5.1 Calculating FT gate

Our objective is to create a gate that reproduces the transformation given by eq. 2.9, separately to even and odd modes.

We first apply a Bean-Splitter Matrix, which we calculate explicitly here. The way to do it is to define the state both in base B^a of space even (odd) modes and B^b of momentum modes:

$$|\psi\rangle_a = \sum_{i_1,i_2=0,1} \psi_{i_1 i_2} \left(c_1^\dagger\right)^{i_1} \left(c_2^\dagger\right)^{i_2} |i_1 i_2\rangle \ , \quad |\psi\rangle_b = \sum_{j_1,j_2=0,1} \psi_{j_1 j_2} \left(b_{k=0}^\dagger\right)^{j_1} \left(b_{k=1}^\dagger\right)^{j_2} |j_1 j_2\rangle \tag{5.1}$$

Now, the next step is to plug $b^\dagger$ as a function of $a^\dagger s$. Then the way to calculate the gate is to compute each coefficient:

$$F^{[(i_1,i_2)(j_1,j_2)]} = \langle i_1 i_2| \left(c_1\right)^{i_1} \left(c_2\right)^{i_2} \left(b_{k=0}^\dagger\right)^{j_1} \left(b_{k=1}^\dagger\right)^{j_2} |j_1 j_2\rangle \tag{5.2}$$

For example, if we calculate for $[(i_1, i_2)\,(j_1, j_2)] = [(0, 0)\,(0, 0)]$ we obtain $F^{[(0,0)(0,0)]} = \langle 00|00\rangle = 1$.

Another example: $F^{[(0,0)(0,1)]} = \langle 00| \left(b_{k=1}^\dagger\right) |01\rangle = 0$.

If we continue we will obtain:

$$\text{BS-gate} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix} \tag{5.3}$$

Then we have to apply a Phase-Shift Gate:

$$Ph\left(i\frac{2\pi k}{n}\right) = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\frac{2\pi k}{n}} \end{bmatrix} \tag{5.4}$$

so we finally obtain eq. 2.10.

5.2 Kyoto Noise Model from IBM

IBM brings us the possibility of using their noise models in Qiskit of their different devices, and in our case we used the one for Kyoto.

As we appreciate the model from IBM does not reproduce well the experimental points, so it is not useful to us to know better the kind of quantum errors that the device has.

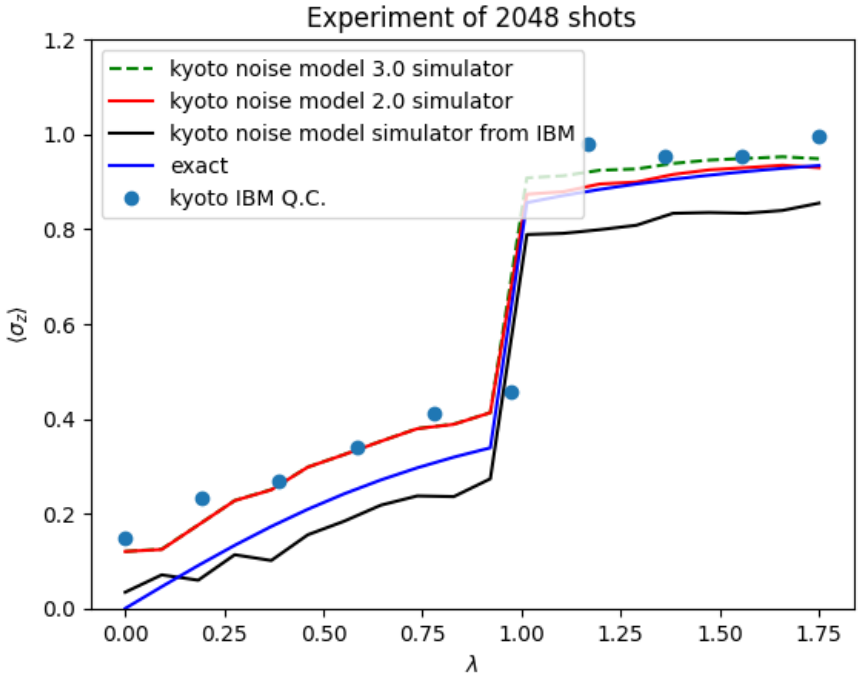


Figure 9: Mean transverse magnetization σ_z as a function of λ. We plot the analytical prediction (blue), the experimental points obtained by IBM-Kyoto, and the diverse noise model simulation predictions. Noise model from IBM plotted in black.